

8-1 · Understand Equations and Their Solutions

Connect & apply · 8 tasks · Prepares for 6.EE.B.5

Use the explanation to discuss strategies. Accept mathematically equivalent answers.

1. FOUNDATION 1

Evaluate $3x + 2$ when $x = 4$.

14

Substitute and multiply first: $3 \times 4 + 2 = 14$.

Skill: Substitute a value into an expression and evaluate

2. FOUNDATION 2

Find the missing number: $7 + 5 = \underline{\quad} + 4$.

8

Both sides must equal 12; $8 + 4 = 12$.

Skill: Read an equals sign as a balance, not as "the answer is"

3. FOUNDATION 3

Write an equation for a number n plus 6 is 15.

$n + 6 = 15$

Is connects two equal quantities, so use an equals sign.

Skill: Translate a sentence into an equation

4. FOUNDATION 4

Does $x = 5$ make $2x + 1 = 11$ true? Explain.

Yes; $2 \times 5 + 1 = 11$

Substitution gives 11 on the left, matching the right side.

Skill: Compare two sides to decide whether they match

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5. CHECK A

Is $x = 4$ a solution of $3x = 12$? Write an equation for "three times a number is 15."

Substitute $x = 4$: $3(4) = 12$ is true, so YES, $x = 4$ is a solution. Equation: $3n = 15$ (or $3x = 15$).

Compare each part of your work with the worked solution.

6. CHECK B

Is $x = 5$ a solution of $4x = 20$? Write an equation for "five times a number is 35."

$4(5) = 20$ is true, so yes. Equation: $5n = 35$.

Compare each part of your work with the worked solution.

7. CHECK C

Is $x = 3$ a solution of $2x + 1 = 8$? Write an equation for "a number increased by 9 is 21."

$2(3) + 1 = 7$, not 8, so no. Equation: $n + 9 = 21$.

Compare each part of your work with the worked solution.

8. TRANSFER

Which of 2, 4, or 6 is a solution of $3x - 2 = 10$?

$x = 4$: $3(4) - 2 = 10$. ✓ The others give 4 and 16.

Explain why your method works and include units when needed.

8-2 · Write and Solve Equations Using Addition or Subtraction

Connect & apply · 8 tasks · Prepares for 6.EE.B.7

Use the explanation to discuss strategies. Accept mathematically equivalent answers.

1. FOUNDATION 1

Find the missing number: $\underline{\quad} + 7 = 19$.

12

Undo adding 7 by subtracting: $19 - 7 = 12$.

Skill: Recall addition and subtraction facts and their inverses

2. FOUNDATION 2

Compute $5.2 - 1.85$.

3.35

Write 5.20 and line up decimal places before subtracting.

Skill: Add and subtract multi-digit numbers, decimals, and fractions

3. FOUNDATION 3

Which operation undoes subtracting 8?

Adding 8

Adding 8 and subtracting 8 are inverse operations.

Skill: Name the inverse of adding or subtracting a number

4. FOUNDATION 4

Check $x = 9$ in $x + 6 = 15$. State the left-side value.

15

Substitute 9: $9 + 6 = 15$, matching the right side.

Skill: Substitute the answer back in to check

8-2 · Write and Solve Equations Using Addition or Subtraction

Connect & apply · 8 tasks · Prepares for 6.EE.B.7

Use the explanation to discuss strategies. Accept mathematically equivalent answers.

5. CHECK A

$m - 11 = 25$, so $m = ?$ What undoes "+ 8"?

$m - 11 = 25 \rightarrow m = 25 + 11 = 36$. What undoes '+ 8': Subtracting 8 (or '- 8').

Compare each part of your work with the worked solution.

6. CHECK B

$m - 14 = 31$, so $m = ?$ What undoes "+ 12"?

$m = 31 + 14 = 45$. Subtracting 12 undoes adding 12.

Compare each part of your work with the worked solution.

7. CHECK C

$y + 18 = 40$, so $y = ?$ What undoes "- 7"?

$y = 40 - 18 = 22$. Adding 7 undoes subtracting 7.

Compare each part of your work with the worked solution.

8. TRANSFER

A jar lost 17 marbles and now holds 48. Write and solve an equation for the original count.

**$m - 17 = 48$, so $m = 65$. Check: $65 - 17 = 48$.
✓**

Explain why your method works and include units when needed.

8-3 · Write and Solve Equations Using Multiplication or Division

Connect & apply · 8 tasks · Prepares for 6.EE.B.7

Use the explanation to discuss strategies. Accept mathematically equivalent answers.

1. FOUNDATION 1

Compute 9×7 .

63

Use $10 \times 7 - 7 = 63$.

Skill: Recall multiplication and division facts

2. FOUNDATION 2

Compute $12.6 \div 3$.

4.2

126 tenths divided by 3 is 42 tenths.

Skill: Divide multi-digit numbers and decimals

3. FOUNDATION 3

Which operation undoes multiplying by 6?

Dividing by 6

Division by the same nonzero factor reverses multiplication.

Skill: Name the inverse of multiplying or dividing by a number

4. FOUNDATION 4

Compute $12 \div \frac{3}{4}$.

16

Multiply by the reciprocal: $12 \times \frac{4}{3} = 16$.

Skill: Divide by a fraction to undo multiplying by one

8-3 · Write and Solve Equations Using Multiplication or Division

Connect & apply · 8 tasks · Prepares for 6.EE.B.7

Use the explanation to discuss strategies. Accept mathematically equivalent answers.

5. CHECK A

$6x = 42$, so $x = ?$ $x \div 5 = 9$, so $x = ?$ What undoes " $\times 4$ "?

$6x = 42 \rightarrow x = 42 \div 6 = 7$. $x \div 5 = 9 \rightarrow x = 9 \times 5 = 45$. What undoes ' $\times 4$ ': Dividing by 4 (or ' $\div 4$ ').

Compare each part of your work with the worked solution.

6. CHECK B

$8x = 56$, so $x = ?$ $x \div 4 = 7$, so $x = ?$ What undoes " $\times 9$ "?

$x = 56 \div 8 = 7$. $x = 7 \times 4 = 28$. Dividing by 9 undoes multiplying by 9.

Compare each part of your work with the worked solution.

7. CHECK C

$5x = 45$, so $x = ?$ $x \div 3 = 12$, so $x = ?$ What undoes " $\div 6$ "?

$x = 45 \div 5 = 9$. $x = 12 \times 3 = 36$. Multiplying by 6 undoes dividing by 6.

Compare each part of your work with the worked solution.

8. TRANSFER

Tickets cost the same amount. Nine tickets cost \$63. Write and solve an equation for one ticket.

$9t = 63$, so $t = \$7$. Check: $9(7) = 63$. ✓

Explain why your method works and include units when needed.

8-4 · Write and Represent Inequalities

Connect & apply · 8 tasks · Prepares for 6.EE.B.8

Use the explanation to discuss strategies. Accept mathematically equivalent answers.

1. FOUNDATION 1

Which symbol means greater than or equal to?

$\geq$

The line under $>$ includes equality at the boundary.

Skill: Use $<$, $>$, $\leq$, and $\geq$ correctly

2. FOUNDATION 2

Write an inequality for a number n that is at least 12.

$n \geq 12$

At least includes 12 and every larger value.

Skill: Translate "at least," "at most," "no more than," "more than"

3. FOUNDATION 3

On a number line marked in ones, what number is two steps left of 1?

-1

Move from 1 to 0, then to -1.

Skill: Place a number on a number line

4. FOUNDATION 4

Does 7 satisfy $n < 5$? Explain.

No; 7 is greater than 5

Substituting 7 gives $7 < 5$, which is false.

Skill: Decide whether a given value makes a comparison true

8-4 · Write and Represent Inequalities

Connect & apply · 8 tasks · Prepares for 6.EE.B.8

Use the explanation to discuss strategies. Accept mathematically equivalent answers.

5. CHECK A

Write an inequality for "you must be at least 48 inches tall." Is 50 a solution?

Inequality: $h \geq 48$. Is 50 a solution? Yes, because $50 \geq 48$ is true.

Compare each part of your work with the worked solution.

6. CHECK B

Write an inequality for "you may bring at most 3 guests." Is 3 a solution?

$g \leq 3$. Yes — $3 \leq 3$ is true.

Compare each part of your work with the worked solution.

7. CHECK C

Write an inequality for "the ride requires more than 40 inches." Is 40 a solution?

$h > 40$. No — $40 > 40$ is false.

Compare each part of your work with the worked solution.

8. TRANSFER

A bus holds no more than 44 riders and already has 31. Write an inequality for how many more can board.

$31 + r \leq 44$, so $r \leq 13$.

Explain why your method works and include units when needed.

8-5 · Understand Inequalities and Their Solutions

Connect & apply · 8 tasks · Prepares for 6.EE.B.5

Use the explanation to discuss strategies. Accept mathematically equivalent answers.

1. FOUNDATION 1

Describe the graph of $x > 3$ on a number line.

Open circle at 3; shade right

Exclude 3 because the symbol is strict; values greater than 3 lie right.

Skill: Mark a boundary point on a number line and shade one direction

2. FOUNDATION 2

For $x \leq 4$, is the circle at 4 open or closed?

Closed

The equality part includes the boundary value 4.

Skill: Choose an open or closed circle from the symbol

3. FOUNDATION 3

Which of 2, 5, 8 satisfy $x \geq 5$?

5 and 8

Both are at least 5; 2 is below the boundary.

Skill: Substitute and test several candidate values

4. FOUNDATION 4

Which is greater: -2 or -6?

-2

-2 is farther right on the number line.

Skill: Compare a value against the boundary number

8-5 · Understand Inequalities and Their Solutions

Connect & apply · 8 tasks · Prepares for 6.EE.B.5

Use the explanation to discuss strategies. Accept mathematically equivalent answers.

5. CHECK A

Graph $x \geq 3$. Open circle or closed? Is 3 itself a solution?

Graph $x \geq 3$: Draw a solid closed circle at 3 and shade an arrow pointing to the right. Circle type: Closed circle. Is 3 a solution? Yes, because $3 \geq 3$ is true.

Compare each part of your work with the worked solution.

6. CHECK B

Graph $x > 5$. Open circle or closed? Is 5 itself a solution?

Open circle at 5, shade to the right. No — $5 > 5$ is false.

Compare each part of your work with the worked solution.

7. CHECK C

Graph $x \leq -2$. Open circle or closed? Is -2 itself a solution?

Closed circle at -2 , shade to the left. Yes — $-2 \leq -2$ is true.

Compare each part of your work with the worked solution.

8. TRANSFER

Which of -1 , 0 , 4 , and 4.5 satisfy $x \geq 4$?

4 and 4.5. The values -1 and 0 are to the left of the boundary.

Explain why your method works and include units when needed.